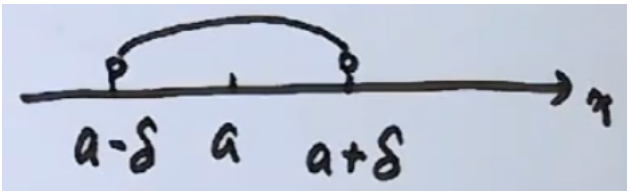
函数及函数的初等特性

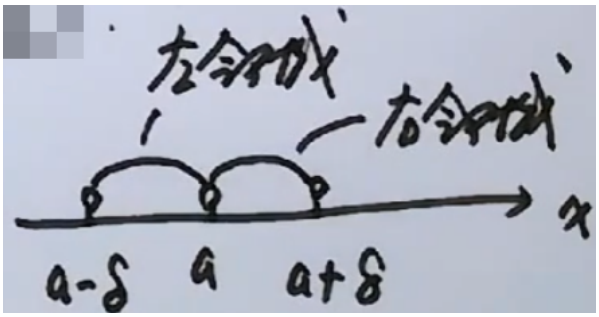
邻域



a的δ邻域

$$a \in R, \delta > 0$$

$$U(a,\delta) = \{x||x-a| < \delta\} = (a-\delta, a+\delta)$$



a的去心δ邻域

$$a \in R, \delta > 0$$

$$\mathring{U}(a,\delta) = \{x|0 < |x-a| < \delta\} = (a-\delta, a) \cup (a, a+\delta)$$

函数

$$R = \{y|y = f(x), x \in D\} - \text{值域}$$

$$y = sgn\ x \begin{cases} -1, x < 0 \\ 0, x = 0 \\ 1, x > 0 \end{cases}$$

$$|a| = a\ sgn\ a$$

$Dirichlet$函数

$$y = D(x) = \begin{cases} 1, x \in \mathbb{Q} \\ 0, x \in \mathbb{R} \backslash \mathbb{Q} \end{cases}$$

取整函数

$$y = [x] - \text{值为不超过}x\text{的最大整数}$$

$$[-3] = -3, [0] = 0, [2] = 2$$

$$[-3.2] = -4, [\sqrt{3}] = 1$$

$$\begin{cases} [x] \le x \\ [x+y] \ne [x] + [y], [-0.2+0.8] = [0.6] = 0, [-0.2] + [0.8] = -1 \\ [x+k] = [x] + k(k \in \mathbb{Z}) \end{cases}$$

反函数

$$y=f(x)(x\in D),R=\{y|y=f(x),x\in D\}$$

$$\text{若}\forall y\in R,\text{由}y=f(x)\text{确定唯一的}x\in D\text{与}x\text{对应}$$

$$x\text{称为}y\text{的函数,记}x=f^{-1}(y)$$

1. $y=f(x)(x\in D)$严格单调 $\Rightarrow\exists$反函数
2. $y=f(x)\Rightarrow x=\Phi(y)$为$y=f(x)$的反函数

$$\text{求函数}y=\ln(x+\sqrt{x^2+1})\text{的反函数.}$$

$$x+\sqrt{x^2+1}=e^y$$

$$\because(-x+\sqrt{x^2+1})(x+\sqrt{x^2+1})=1$$

$$\therefore -x+\sqrt{x^2+1}=\frac{1}{x+\sqrt{x^2+1}}=e^{-y}$$

$$\Rightarrow\text{反函数为}x=\frac{e^y-e^{-y}}{2}$$

基本初等函数

1. $y=x^a,y=x^3(-\infty<x<+\infty),y=\sqrt{x}(0\le x<+\infty)$
2. $y=a^x(a>0\text{且}a\ne 1)(-\infty<x<+\infty)$
3. $y=\log_a x(a>0\text{且}a\ne 1)(0<x<+\infty)$
4. $\begin{cases}\sin x,\cos x:(-\infty<x<+\infty)\\ \sec x=\frac{1}{\cos x},\tan x:\{x|x\in\mathbb{R},x\ne k\pi+\frac{\pi}{2}(k\in\mathbb{Z})\}\\ \csc x=\frac{1}{\sin x},\cot x:\{x|x\in\mathbb{R},x\ne k\pi(k\in\mathbb{Z})\}\end{cases}$
5. $\begin{cases}\arcsin x,\arccos x:(-1\le x\le 1)\\ \arctan x,arccot\ x:(-\infty<x<+\infty)\end{cases}$

初等函数

$$\text{由}\begin{cases}\text{常数}\\ \text{基本初等函数}\end{cases}\text{经过}\begin{cases}\text{四则}\\ \text{复合}\end{cases}\text{而成的式子}$$

$$\text{称为初等函数}$$

初等特性

有界性

$$\text{对}\{a_n\},\text{若}\exists M>0,\text{使}\forall n,\text{有}|a_n|\le M(-M\le a_n\le M),\text{称}\{a_n\}\text{有界}$$

$$\text{若}\exists M_1,\forall n,a_n\ge M_1,\{a_n\}\text{有下界}$$

$$\text{若}\exists M_2,\forall n,a_n\le M_2,\{a_n\}\text{有上界}$$

$$|a_n|\le 2\Rightarrow\begin{cases}a_n\ge -2\\ a_n\le 2\end{cases}$$

$$\begin{cases}a_n\ge -3\\ a_n\le 5\end{cases}\Rightarrow|a_n|\le 5$$

$$\{a_n\}\text{有界}\Leftrightarrow\{a_n\}\text{有上、下界}$$

$$y=f(x)(x\in D)$$

$$\text{若}\exists M>0,\text{对}\forall x\in D,|f(x)|\le M$$
$$\text{称}f(x)\text{在}D\text{上有界}$$
$$\text{若}\exists M_1,\text{对}\forall x\in D,|f(x)|\ge M_1$$
$$\text{称}f(x)\text{在}D\text{上有下界}$$
$$\text{若}\exists M_2,\text{对}\forall x\in D,|f(x)|\le M_2$$
$$\text{称}f(x)\text{在}D\text{上有上界}$$

$$f(x)\text{在}D\text{上有界}\Leftrightarrow f(x)\text{在}D\text{上有上、下界}$$

奇偶性

$$y=f(x)(x\in D),D\text{关于原点对称}$$

$$\text{讨论函数}f(x)=\ln(x+\sqrt{x^2+1})\text{的奇偶性.}$$

$$x\in(-\infty,+\infty)$$
$$f(-x)=\ln(-x+\sqrt{x^2+1})=\ln\frac{1}{x+\sqrt{x^2+1}}=-\ln(x+\sqrt{x^2+1})=-f(x)$$
$$\therefore f(x)=\ln(x+\sqrt{x^2+1})\text{为奇函数}$$

设函数$f(x)$在D上有定义，且D关于原点对称，证明：函数$f(x)$可表示为一个奇函数和一个偶函数之和.

$$f(x)=\frac{f(x)+f(-x)}{2}+\frac{f(x)-f(-x)}{2}=F(x)+G(x)$$
$$F(-x)=\frac{f(-x)+f(x)}{2}=F(x)$$
$$G(-x)=\frac{f(-x)-f(x)}{2}=-\frac{f(x)-f(-x)}{2}=-G(x)$$

单调性

周期性